

❏ 1. Shutdown and Installation Concepts

❏ Shutdown Procedure:

Shutting down Linux safely ensures data integrity and avoids filesystem corruption. Common commands:

Command	Description
shutdown -h now	Halt the system immediately
shutdown -r now	Reboot the system immediately
reboot	Alias for shutdown -r now
halt	Stops all CPU functions but may not power off
poweroff	Turns off the system
init 0	Changes runlevel to 0 (halt)
init 6	Changes runlevel to 6 (reboot)
systemctl poweroff	Used in systemd to power off
systemctl reboot	Used in systemd to reboot

❏ Scheduling Shutdown:

```
bash
```

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```
shutdown -h +10 "System will shut down in 10 minutes"
```

Cancel scheduled shutdown:

```
bash
```

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```
shutdown -c
```

❏ 2. Kickstart Configuration & Customization

❏ What is Kickstart?

Kickstart automates Linux installation using a configuration file with predefined settings.

❏ File Location:

A Kickstart file is typically stored as ks.cfg and can be provided via:

- USB/CD-ROM
- Network (NFS, HTTP, FTP)

❏ Important Sections of a Kickstart File:

1.
Installation Settings

nginx

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```
lang en_US keyboard us timezone Asia/Kolkata rootpw password123
```

2.

Partitioning

css

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```
clearpart --all --initlabel autopart
```

3.

Bootloader

lua

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bootloader --location=mbr
```

4.

Network Configuration

```
lua
```

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```
network --bootproto=dhcp --hostname=server1.local
```

5.

Package Selection

```
perl
```

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```
%packages @core httpd vim %end
```

6.

Post Installation Scripts

```
lua
```

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```
%post echo "Installation complete" >> /var/log/kickstart.log %end
```

☒ Tools:

- system-config-kickstart (GUI)
- Generate Kickstart via:

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anaconda-ks.cfg

❏ 3. Deployment using Kickstart

❏ Steps for Kickstart Deployment:

1.
Create ks.cfg file
2.
Place the file in a known location (USB, HTTP server)
3.
Boot from installation media

4.

Specify the Kickstart file location:

nginx

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```
linux ks=http://192.168.1.10/ks.cfg
```

5.

Installer reads config and proceeds without prompts

☒ **Advanced Kickstart Options:**

- RAID setup
- LVM configuration

- Custom partitioning
 - Script execution (e.g., software installation, post-config)
-

❏ 4. User Administration

❏ Creating Users:

```
bash
```

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```
useradd alice passwd alice
```

❏ Modifying Users:

bash

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```
usermod -aG wheel alice    # Add to group usermod -s /bin/bash alice # Change shell usermod -L alice    # Lock account
```

❏ Deleting Users:

bash

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```
userdel alice userdel -r alice # Remove home directory too
```

❏ Viewing User Info:

- /etc/passwd: Basic user info

- /etc/shadow: Passwords (hashed)
- /etc/group: Group info
- id alice: UID, GID, groups

📄 Group Management:

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```
groupadd dev gpasswd -a alice dev
```

📄 Sudo Access:

- Add user to wheel or sudo group

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Or configure `/etc/sudoers` using visudo